The action of a componentwise map on a pair path agrees with the pair path of the component actions

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled thm:ap-prod (source lines 1378-1381); the proof span 1382-1389 is the author's proof environment, retained line for line.

Editorial note. The pairpath notation used by the statement is the author's name for the inverse from the product path characterization, covered by the standard prerequisites; the statement and proof cite no external labels.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: naturality of the pair path construction under componentwise maps. Only the counted theorem and its author proof are retained; the author's set-up prose for the componentwise map f is background.

Allowed standard prerequisites (editorial).

- Cartesian products, componentwise maps f sending a pair to the pair of component images.
- The pair path construction: the inverse map from pairs of component paths to paths in the product, named pairpath by the author.
- The action of a function on paths (ap) with its definitional behaviour on reflexivity.
- Path induction and product induction with their conversion rules.

Theorem 0.0.1. *In the above situation, given $x, y : A \times B$ and $p : \mathsf{pr}_1 x = \mathsf{pr}_1 y$ and $q : \mathsf{pr}_2 x = \mathsf{pr}_2 y$, we have*

$$f(\mathsf{pair}^{=}(p, q)) =_{(f(x)=f(y))} \mathsf{pair}^{=}(g(p), h(q)).$$

Proof. Note first that the above equation is well-typed. On the one hand, since $\mathsf{pair}^{=}(p, q) : x = y$ we have $f(\mathsf{pair}^{=}(p, q)) : f(x) = f(y)$. On the other hand, since $\mathsf{pr}_1(f(x)) \equiv g(\mathsf{pr}_1 x)$ and $\mathsf{pr}_2(f(x)) \equiv h(\mathsf{pr}_2 x)$, we also have $\mathsf{pair}^{=}(g(p), h(q)) : f(x) = f(y)$.

Now, by induction, we may assume $x \equiv (a, b)$ and $y \equiv (a', b')$, in which case we have $p : a = a'$ and $q : b = b'$. Thus, by path induction, we may assume p and q are reflexivity, in which case the desired equation holds judgmentally. □